Unit 1
Lesson 4: Historical Developments of Hardware and Operating System

What is Operating System ?

A program that acts as an intermediary between a user of a computer and the computer hardware.

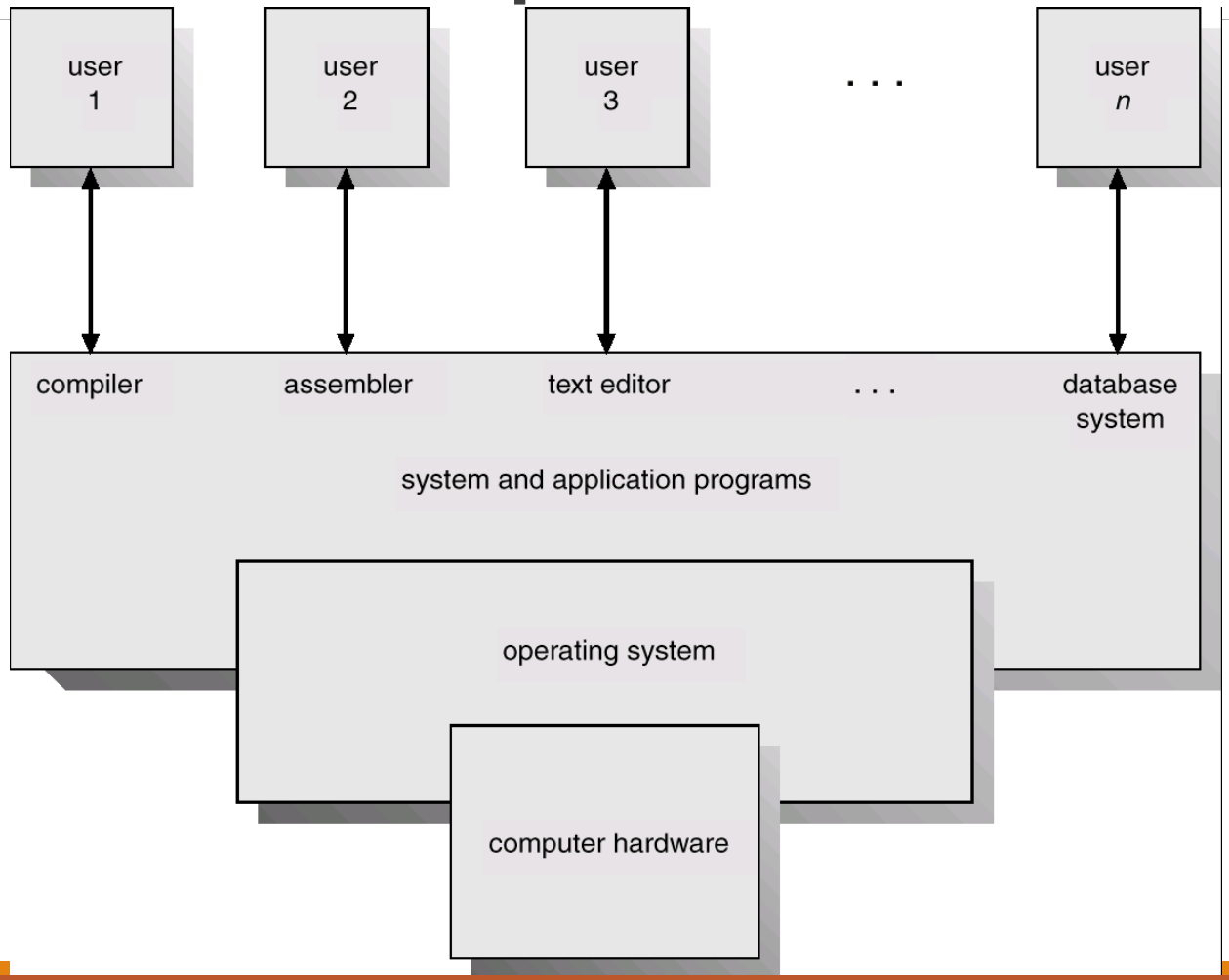
Operating system goals:

- A program that controls the execution of application programs
- Make the computer system convenient to use.
- Use the computer hardware in an efficient manner.

Computer System Components

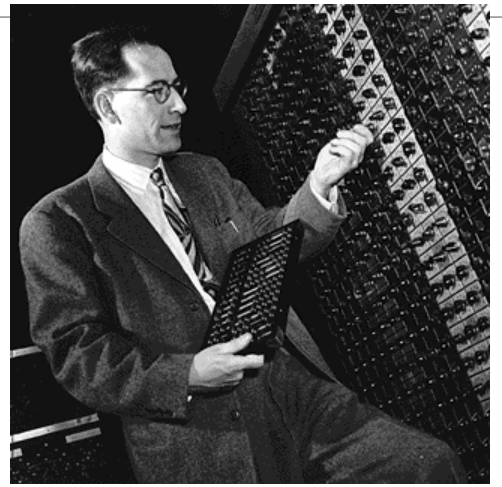
1. **Hardware** – provides basic computing resources (CPU, memory, I/O devices).
2. **Operating system** – controls and coordinates the use of the hardware among the various application programs for the various users.
3. **Applications programs** – define the ways in which the system resources are used to solve the computing problems of the users (compilers, database systems, video games, business programs).
4. **Users** (people)

Abstract View of System Components



A bit of history

The First Generation (1945-55) – Vacuum Tubes and Plugboards







History (ctd.)

The Second Generation (1955-65)
Transistors and Batch Systems

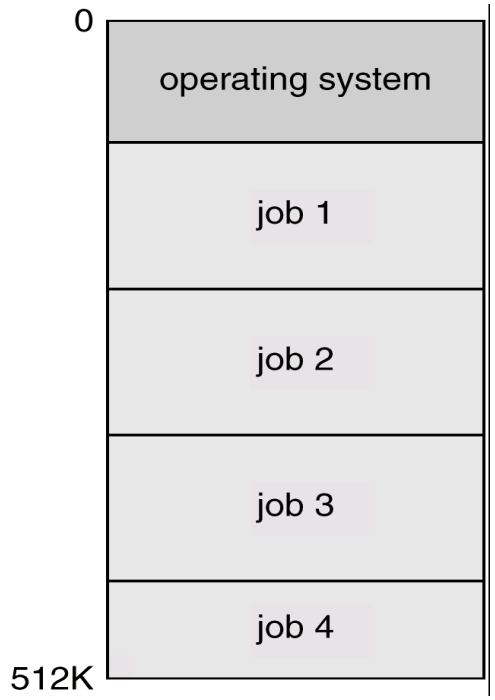
- Using FORTRAN and assembly language
- OS – FMS , IBSYS
  (for IBM7094), Scope



History (ctd.)

The Third Generation (1965-1980) –
Multiprogramming and Spooling
The Primary development : PDPs computer and UNIX os

- Several jobs are kept in main memory at the same time, and the CPU is multiplexed among them.
- Spooling - While executing one job, the OS:
  - Reads next job from card reader into a storage area on the disk (job queue).
  - Outputs printout of previous job from disk to printer.
  - *Job pool* – data structure that allows the OS to select which job to run next in order to increase CPU utilization.



OS Features Needed for Multiprogramming

- I/O routine supplied by the system.
- Memory management – the system must allocate the memory to several jobs.
- CPU scheduling – the system must choose among several jobs ready to run.

Example

	JOB1	JOB2	JOB3
Type of job	Heavy compute	Heavy I/O	Heavy I/O
Duration	5 min.	15 min.	10 min.
Memory required	50K	100 K	80 K
Need disk?	No	No	Yes
Need terminal	No	Yes	No
Need printer?	No	No	Yes

Effects of Multiprogramming

	Uniprogramming	Multiprogramming
Processor use	17%	33%
Memory use	30%	67%
Disk use	33%	67%
Printer use	33%	67%
Elapsed time	30 min.	15 min.
Throughput rate	6 jobs/hr	12 jobs/hr
Mean response time	18 min.	10 min.

Time sharing advantages

Using multiprogramming (multi-tasking) to handle multiple interactive jobs

Processor's time is shared among multiple users

Multiple users simultaneously access the system through terminals

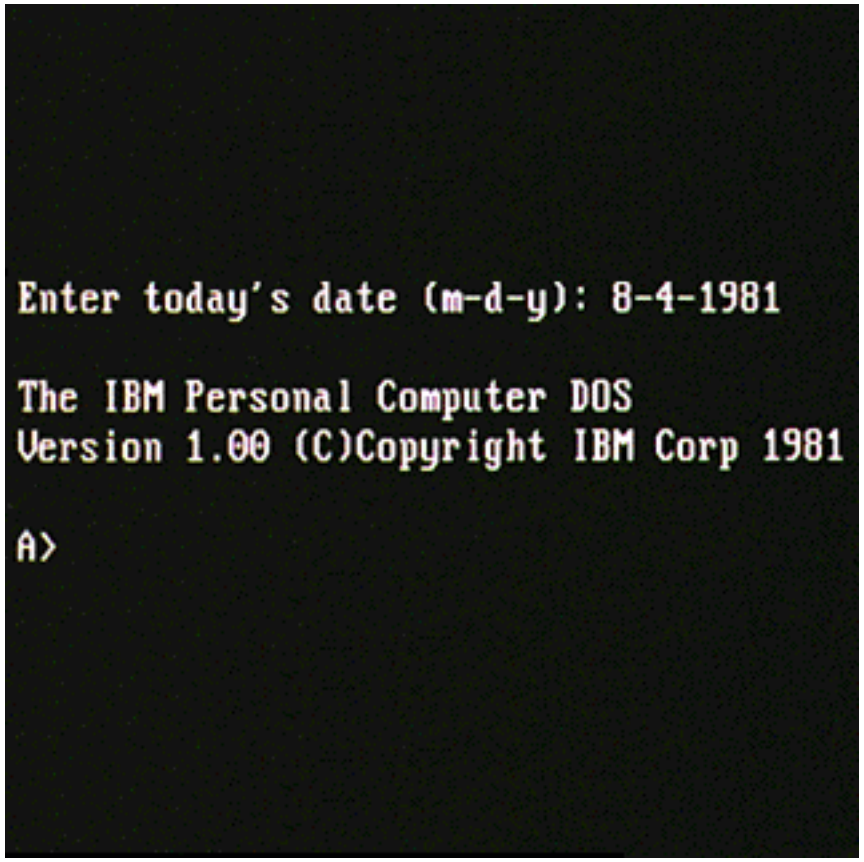
The objective is to minimize response time to the user commands

History (cont.)

The Fourth Generation(1980 – Present) – Personal Computers

- *Personal computers* – computer system dedicated to a single user.
- I/O devices – keyboards, mice, display screens, small printers.
- User convenience and responsiveness.
- Can adopt technology developed for larger operating system' often individuals have sole use of computer and do not need advanced CPU utilization of protection features

Windows - History



- **DOS 1.00: 4000 lines of assembly language**

Windows - History

DOS 2.0 in 1983, supported a hard disk, one directory, max 64 files

DOS 3.0 ran on Intel's 286 processor, memory protection features and hard disk

DOS 3.1, 1984, supported networking

DOS 3.3, 1987, supported 386 processor

Windows 3.0, 1990, PC Graphical User Interface (GUI) for the 1st time

Windows - History

Windows NT developed from scratch to provide a multitasking single user environment. Version 3.1, 1993.

Windows 95-98 :
Multitasking single user – not full 32bit

Win NT 4.0, 1996 – win95 GUI
16 million lines of C code

History of Operating System

Generation	Year	Electronic Device Used	Operating System Device Used
First Generation	1945 - 1955	Vacuum Tubes	Plug Boards
Second Generation	1955 - 1965	Transistors	Batch Systems
Third Generation	1965-1980	Integrated Circuits	Multiprogramming
Fourth Generation	1980 - Present	Large scale Integration Devices	Personal Computers

Characteristics of Operating System

Memory Management – Keeps track of the primary memory, i.e. what part of it is in use by whom, what part is not in use, etc. and allocates the memory when a process or program requests it.

Processor Management – Allocates the processor (CPU) to a process and deallocates the processor when it is no longer required.

Characteristics of Operating System

Device Management – Keeps track of all the devices. This is also called I/O controller that decides which process gets the device, when, and for how much time.

File Management – Allocates and de-allocates the resources and decides who gets the resources.

Characteristics of Operating System

Security – Prevents unauthorized access to programs and data by means of passwords and other similar techniques.

Job Accounting – Keeps track of time and resources used by various jobs and/or users.

Control Over System Performance – Records delays between the request for a service and from the system.

Characteristics of Operating System

Interaction with the Operators – Interaction may take place via the console of the computer in the form of instructions. The Operating System acknowledges the same, does the corresponding action, and informs the operation by a display screen.

Error-detecting Aids – Production of dumps, traces, error messages, and other debugging and error-detecting methods.

Characteristics of Operating System

Coordination Between Other Software and Users – Coordination and assignment of compilers, interpreters, assemblers, and other software to the various users of the computer systems.

Types of Operating System

Batch Operating Systems

In these, the jobs with similar needs were batched up and executed together. Then the results of these jobs were sent back to the respective programmer.

Types of Operating System

Time Sharing Operating Systems

This type of operating system allows multiple users to share the system simultaneously. Each user is provided a time slice and the processor switches rapidly among the users according to it. Because of this, each user believes that they are the only ones using the system.

Types of Operating System

Multiprocessor Operating Systems

These operating systems have multiple processors working together in parallel. These processors share the system resources such as bus, clock, memory etc.

Distributed Operating Systems

These systems manage a set of different computers that are spread across a network. So, the distributed operating systems allow the distributed systems to share resources and communicate efficiently.

Types of Operating System

Real-Time Operating Systems

There are strict time specifications in real time operating systems. So they are used in scientific projects, banks, robotics etc. where time bounds are important.

I remember that…

Resources:

Text Books :

- Applied Operating System Concepts – Silberschatz/Galvin/Gagne
  http://www.bell-labs.com/topic/books/aos-book
- Modern Operating Systems – Tanenbaum
- Linux Programming – Stones/Matthew
- Advanced Windows – Richter